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# applications - ReCount paper
# september 23, 2011
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library(Biobase)

##### application 1

load(url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/bottomly_eset.RData"))
library(preprocessCore)

bott.dat <- exprs(bottomly.eset)
strain <- phenoData(bottomly.eset)@data$strain
bott.dat.norm75 <- matrix(NA,nrow=36536,ncol=21)
for(i in 1:21) {
  b.zeros <- which(as.vector(bott.dat[,i])==0)
  b.pct75s <- quantile(as.vector(bott.dat[-b.zeros,i]),0.75)
  bott.dat.norm75[,i] <- bott.dat[,i]/b.pct75s }

bott.dat.qnorm <- normalize.quantiles(bott.dat)

# test for differential expression on both normalization schemes:
library(genefilter)
difftest.75 <- rowFtests(bott.dat.norm75,fac=strain, var.equal=FALSE)
difftest.q <- rowFtests(bott.dat.qnorm,fac=strain,var.equal=FALSE)
p75 <- p.adjust(difftest.75$p.value,"BH") #multiple-test-corrected p-values
pq <- p.adjust(difftest.q$p.value,"BH")
length(which(p75<0.05 & pq>=0.05)) #35
length(which(p75>=0.05 & pq<0.05)) #177
length(which(p75<0.05 & pq<0.05))
length(which(p75>=0.05 & pq>=0.05))
sum(is.na(p75)) #23,697
sum(is.na(pq)) #23,697

# venn diagram to visualize:

library(VennDiagram)
venn.diagram(
  list("75th-percentile"=which(p75<0.05),"quantile"=which(pq<0.05)),
  filename="venn.tiff",
  scaled=TRUE,
  lwd=c(1,1),
  col=c("black","black"),
  fill=c("red","blue"),
  alpha=c(0.5,0.5),
  fontfamily="sans",cat.fontfamily="sans",
  cat.col=c("red","blue"),cat.cex=c(1,0.8),cat.pos=c(-130,137)
)

# LOTS of overlap.

##### application 2

### comparing cheung & montgomery bio reps
load(url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/cheung_eset.RData"))
load(url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/montpick_eset.RData"))

cheung.id <- phenoData(cheung.eset)@data$sample.id
mont.id <-
phenoData(montpick.eset)@data$sample.id[which(phenoData(montpick.eset)@data$study=="Montgomery")]
intersect(cheung.id,mont.id) #29 of the same samples assayed in both studies

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expdata <- cbind(exprs(cheung.eset),
  exprs(montpick.eset)[,which(phenoData(montpick.eset)@data$study=="Montgomery")])
expdata <- as.matrix(expdata)

#normalize:
n <- length(cheung.id)+length(mont.id)
expdata.norm <- matrix(NA,nrow=52580,ncol=n)
pct75s <- NULL
zeros <- NULL
for(i in 1:n) { zeros[[i]] <- which(as.vector(expdata[,i])==0) }
for(i in 1:n) { pct75s[i] <- quantile(as.vector(expdata[-zeros[[i]],i]),0.75) }
for(i in 1:n) { expdata.norm[,i] <- expdata[,i]/pct75s[i] }

study <- c(rep("cheung",length(cheung.id)),rep("montgomery",length(mont.id)))
colnames(expdata.norm) <- c(as.character(cheung.id),as.character(mont.id))
matched.temp <- expdata.norm[,which(colnames(expdata.norm) %in%
intersect(cheung.id,mont.id))]
matched <- matrix(NA,nrow=52580,ncol=58)
for(i in 1:29) { matched[,i] <- matched.temp[,i] }
for(i in 30:58) {
  matched[,i] <- matched.temp[,30:58][,which(colnames(matched.temp)
[30:58]==colnames(matched.temp)[i])] }

# now "matched" contains the (normalized) data for cheung and montgomery, only the
intersecting samples,
# in the *same order* (i.e. first 29 columns, 2nd 29 columns are the same samples in same
order)

# now do paired t-tests to see which genes are differentially expressed in the different
studies
paired.results <- matrix(NA,nrow=52580,ncol=3)
colnames(paired.results) <- c("gene","test.statistic","p.value")
for(i in 1:52580){
  paired.results[i,1] <- featureNames(cheung.eset)[i]
  temp <- t.test(matched[i,1:29],matched[i,30:58],paired=TRUE)
  paired.results[i,2] <- temp$statistic
  paired.results[i,3] <- temp$p.value }

bh.p <- p.adjust(paired.results[,3],"BH")
length(which(bh.p<0.05))/length(bh.p) #0.069 -- percentage of genes DE between studies
length(which(bh.p<0.05))/length(bh.p[which(is.na(bh.p)==FALSE)]) #0.283 -- percentage of
testable genes DE between studies
## FIGURE 1
hist(bh.p,col="gray70",main="",xlab="Benjamini-Hochberg-corrected p-value")
length(which(is.na(bh.p)==TRUE))/length(bh.p) #0.756

### comparing montgomery and pickrell studies

# normalize
mpdata <- exprs(montpick.eset)
n <- dim(mpdata)[2]
mpdata.norm <- matrix(NA,nrow=52580,ncol=n)
pct75s <- NULL
zeros <- NULL
for(i in 1:n) { zeros[[i]] <- which(as.vector(mpdata[,i])==0) }
for(i in 1:n) { pct75s[i] <- quantile(as.vector(mpdata[-zeros[[i]],i]),0.75) }
for(i in 1:n) { mpdata.norm[,i] <- mpdata[,i]/pct75s[i] }

# t-tests between studies:
library(genefilter)
difftest <- rowttests(mpdata.norm,fac=as.factor(phenoData(montpick.eset)@data$population))
bh.mp.p <- p.adjust(difftest$p.value,"BH")
length(which(bh.mp.p<0.05))/length(bh.mp.p) #0.089 -- % genes DE between ethnicities
length(which(bh.mp.p<0.05))/length(bh.mp.p[which(is.na(bh.mp.p)==FALSE)]) #0.360 -- % DE
testable genes

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## FIGURE 2
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hist(bh.mp.p,main="",xlab="Benjamini-Hochberg-corrected p-value",col="gray70")  
length(which(is.na(bh.mp.p)==TRUE))/length(bh.mp.p) #0.753
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